



Research paper

Boundary effects of anisotropic medium on cloaking under shallow-water waves

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ARTICLE INFO

Handling Editor: Prof. A.I. Incecik

Keywords:

Cloaking
Space transformation method
Shallow-water wave
anisotropic medium
Offshore structure
Multibodies

ABSTRACT

The cloaking of offshore structures exposed to water waves can effectively protect them from hydrodynamics. In this study, an anisotropic medium designed by applying the space transformation method was used to cloak offshore structures exposed to shallow-water waves. By designing a theoretical cloaking space, the wave path could be controlled to bypass a certain area and cloak the structure within the wave field. Based on the form invariance of the shallow-water wave-governing equation, this unique wave path can be physically created using an anisotropic medium that includes water depth and gravity acceleration. First, an anisotropic medium was designed to cloak a single cylinder. The effects of the no-flux inner boundary conditions of the anisotropic medium on cloaking were investigated. The total- and scattered-wave distributions around the structure were calculated and analyzed. The scattering attenuation coefficient was used to measure the cloaking effect. In addition, a numerical truncation method for the medium parameters near the inner boundary was proposed and investigated to analyze the singularity of the medium parameters near the inner boundary. Moreover, an anisotropic medium was introduced to cloak multibody structures in water waves using simplified parameters. The results indicated that the inner area could also be protected without a no-flux boundary. Cloaking was effective in a simplified anisotropic medium. An anisotropic medium with simplified parameters can successfully cloak multibody structures exposed to shallow-water waves.

1. Introduction

The active manipulation of water waves based on wave properties, including refraction, reflection, and resonance, has attracted attention and can be applied to cloaks (Farhat et al., 2008; Porter, 2011), concentrators (Li et al., 2018; Zhang et al., 2020a), rotators (Chen et al., 2009), shifters (Berraquero et al., 2013), carpets (Dupont et al., 2015; Wang et al., 2017), waveguides (Iida and Kashiwagi, 2017), and wavelength modulators (Zhang et al., 2023a). Among these applications, cloaks can hydrodynamically protect offshore structures from waves. Large wave loads can damage structures (Sarkar and Chanda, 2022; Saha et al., 2023; Chanda et al., 2022). Generally, a cloaked structure does not generate scattered waves in the far field.

Based on the wave property, the scattering cancellation method (SCM), space transformation method (STM), and other methods have been proposed for designing cloaked structures in water waves. An SCM-

based cloak diminishes scattered waves in the far field using the wave resonance (Zhang et al., 2020b). Porter (2011) first used the SCM to diminish the scattered water waves of a vertical cylinder by optimizing the topography around the cylinder. In addition to designing topography, Newman (2013) introduced an array of small cylinders to cloak a fixed truncated cylinder in water. Iida et al. (2023) cloaked a bottom-mounted cylinder using a floating elastic composite plate. The energy of the scattered waves was minimized by designing the flexural rigidity and mass of the plate. Numerical and experimental studies (Iida et al., 2014, 2016; Read et al., 2016; Zhang et al., 2017, 2019, 2023b) on the hydrodynamic protection of offshore structures by cloaking have shown that a few wave drift forces act on the SCM-based cloaked structure. In addition to the SCM, the STM is a more flexible approach because waves can be directly controlled to avoid the space where the objects are located. The wave path can be controlled by designing a theoretical cloaking space to bypass a specific area. Based on the form

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<https://doi.org/10.1016/j.oceaneng.2023.116485>

Received 22 August 2023; Received in revised form 26 November 2023; Accepted 28 November 2023

Available online 5 December 2023

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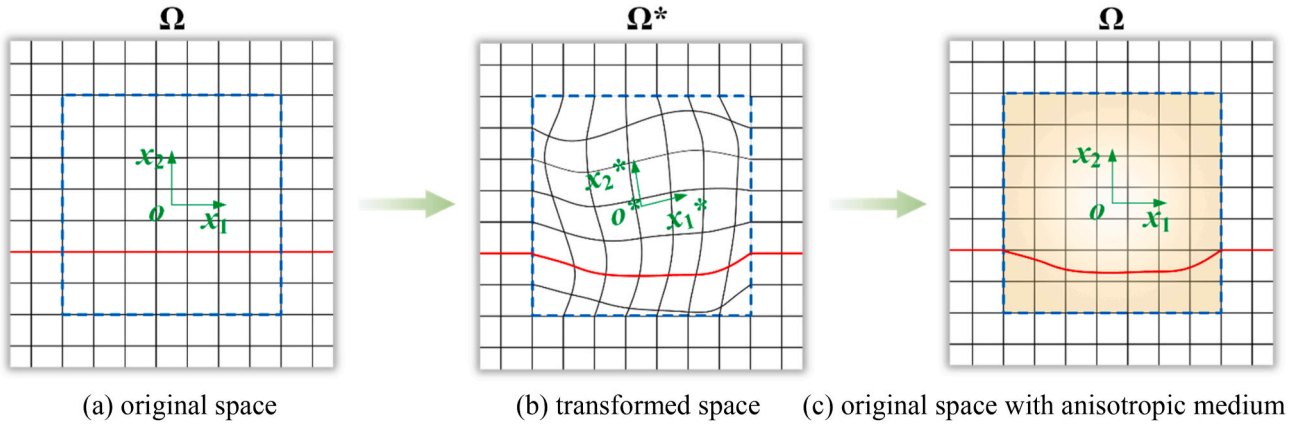


Fig. 1. Original space and transformed space in STM.

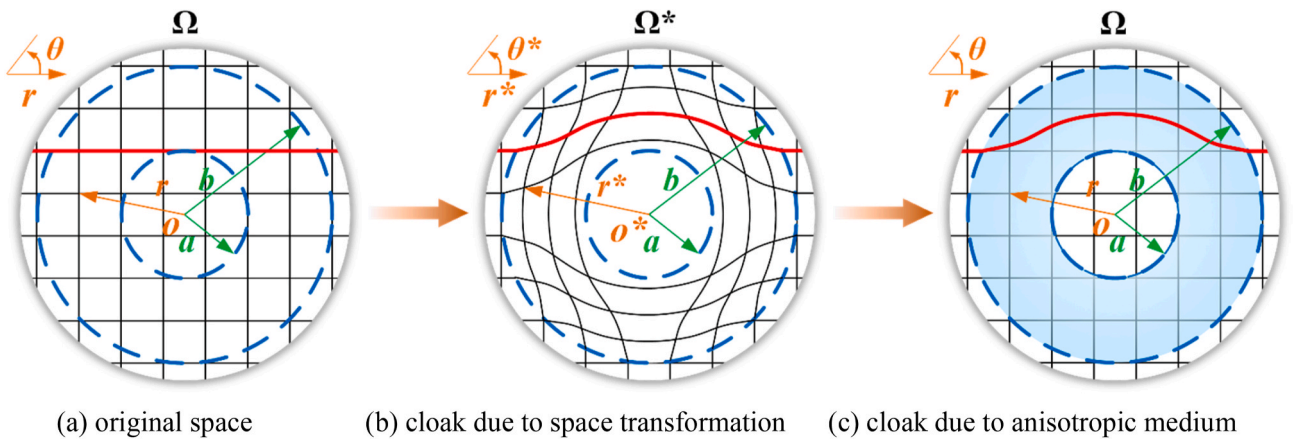


Fig. 2. Design of an annular cloak according to the STM.

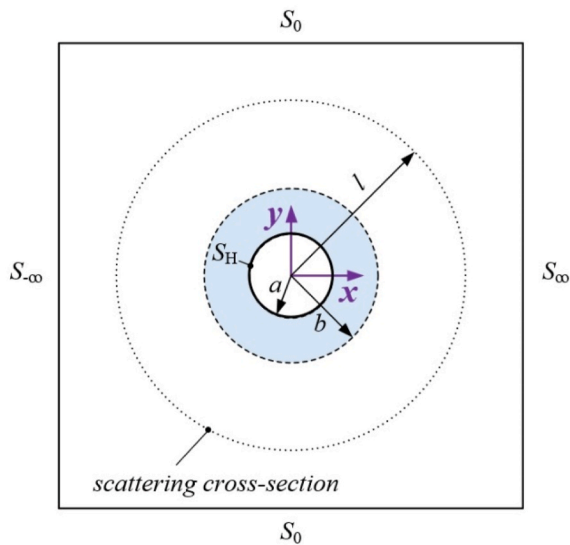


Fig. 3. Model set of annular cloak.

invariance of the governing equation of shallow-water waves, this unique wave path can be physically created using an anisotropic medium, including water depth and gravity acceleration. This implies that waves can be manipulated using a special material, that is, an anisotropic medium. Farhat et al. (2008) first introduced the STM to analyze

the surface waves of methoxynonafluorobutane. The surface waves were controlled to prevent a certain space cloaking the body in the liquid by designing an anisotropic liquid viscosity. Zareei and Alam (2015) proposed a nonlinear space transformation of the shallow-water-governing equation, in which only the anisotropic water depth is required. Iida and Kashiwagi (2018) developed an anisotropic water channel network using the STM to cloak a body in shallow water. The numerical results showed that the proposed anisotropic network functions satisfactorily for long waves. Zhang et al. (2022) investigated the physical mechanism of an STM-based cloak using the superposition principle. The results indicated that the scattered waves of the anisotropic medium changed the wave path to redistribute the wave field.

The STM-based cloak has been theoretically and numerically validated in the design of self-protected offshore structures exposed to water waves. However, some issues remain unclear, particularly the effects of the inner boundary of the anisotropic medium. For example, existing STM-based cloaks cannot be used to protect a freely moving object because a no-flux boundary of the anisotropic medium should be applied. In addition, the designed STM-based cloaking metamaterial has a large vertical size owing to the singularity of the anisotropic medium parameters near the inner boundary. Thus, the subjects of this study are the theoretical and numerical investigations of the STM-based cloak: 1) the effects of the inner no-flux boundary of the anisotropic medium on cloaking, 2) a simplified method of the medium parameters for the singularity problem of the anisotropic parameters near the inner boundary, and 3) cloaking multibodies in water waves using an anisotropic medium. First, the theory relevant to the STM and the design of the anisotropic medium of an annular cloak were introduced. The STM-based cloak was validated by investigating the wave distribution.

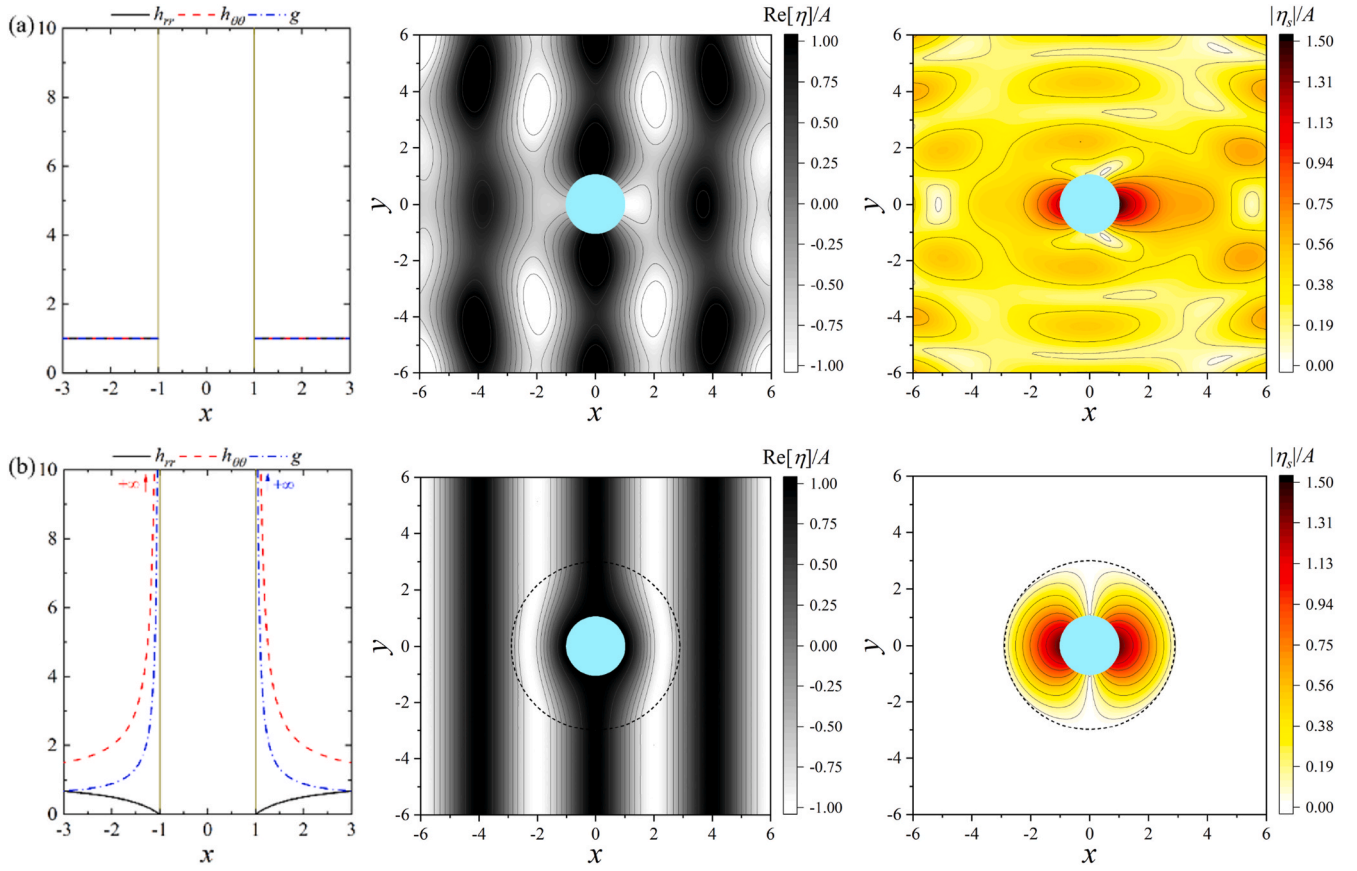


Fig. 4. Anisotropic medium parameters (h_{rr} , $h_{\theta\theta}$, and g), total-wave distribution ($\text{Re}[\eta]/A$), and scattered-wave distribution ($|\eta_s|/A$): (a) single cylinder without anisotropic medium; (b) cloaked cylinder with anisotropic medium.

Moreover, various inner-boundary locations were considered to analyze their effects on cloaking. Furthermore, different simplified cases of anisotropic medium parameters were considered and analyzed to demonstrate their cloaking effects. Finally, an anisotropic medium was introduced to cloak a four-column structure. Theoretically designed anisotropic media can be physically modeled using metamaterials.

2. Theory

2.1. Water-wave governing equation and the STM

In shallow water, under the assumption of an incompressible and inviscid fluid with irrotational motion, the surface waves can be expressed in a two-dimensional form as follows (Mei et al., 2005),

$$\frac{\partial^2 \zeta}{\partial t^2} - \nabla \cdot (g_0 h_0 \nabla \zeta) = 0 \quad (1)$$

where $\zeta(x_1, x_2; t)$ is the wave elevation; g_0 is the original gravity acceleration; h_0 is the original water depth.

Assuming the wave is harmonic, $\zeta = \eta(x_1, x_2)e^{-i\omega t}$, η is the spatial wave elevation, $\eta > 0$ means the water surface is same high as the calm water surface, $\eta < 0$ means the water surface is lower than the calm water surface, $\eta > 0$ means the water surface is higher than the calm water surface, then the following equation can be obtained,

$$\nabla \cdot (h_0 \nabla \eta) + \frac{\omega^2}{g_0} \eta = 0 \quad (2)$$

Equation (2) is the well-known Helmholtz equation, which is form invariance under a space transformation (Chen et al., 2009; Berraquero et al., 2013; Dupont et al., 2015; Wang et al., 2017; Iida and Kashiwagi,

2017, 2018; Zhang et al., 2022, 2023a). Considering a space transformation $\Omega^* = f(\Omega)$ of orthogonal coordinates with the Jacobian Matrix $J = \partial x_i^* / \partial x_j$, where $\Omega(x_1, x_2)$ and $\Omega^*(x_1^*, x_2^*)$ are the original space and the transformed space, respectively, as shown in Fig. 1 (a) and Fig. 1 (b). The new Helmholtz equation under the space transformation can be obtained as follows,

$$\nabla^* \cdot (\tilde{h}^* \nabla^* \eta) + \frac{\omega^2}{g^*} \eta = 0 \quad (3)$$

where $\tilde{h}^* = \frac{JJ^T}{|J|} h_0 = \tilde{h} h_0$, $g^* = |J| g_0 = g |J|$.

$$J = \begin{bmatrix} \frac{\partial x_1^*}{\partial x_1} & \frac{\partial x_1^*}{\partial x_2} \\ \frac{\partial x_2^*}{\partial x_1} & \frac{\partial x_2^*}{\partial x_2} \end{bmatrix} \quad (4)$$

Comparing Eq. (2) with Eq. (3), it can be found that their forms are the same due to the space components ($JJ^T/|J|$ and $1/|J|$) being absorbed into the anisotropic medium parameters ($\tilde{h}^*$ and g^*). That means the same wave distribution in the space Ω^* can be realized by redesigning the medium parameters in the space Ω , as shown in Fig. 1 (c).

Here, $\tilde{h}$ and g are defined as space scale factors. Note that $\tilde{h}$ is a tensor matrix, which can be expressed in a cylindrical coordinate system as follows,

$$\tilde{h} = \begin{bmatrix} h_{rr} & h_{r\theta} \\ h_{\theta r} & h_{\theta\theta} \end{bmatrix} = \frac{JJ^T}{|J|}, g = |J| \quad (5)$$

2.2. Design of the anisotropic medium of an annular cloak

An annular cloak is considered to accommodate different incident

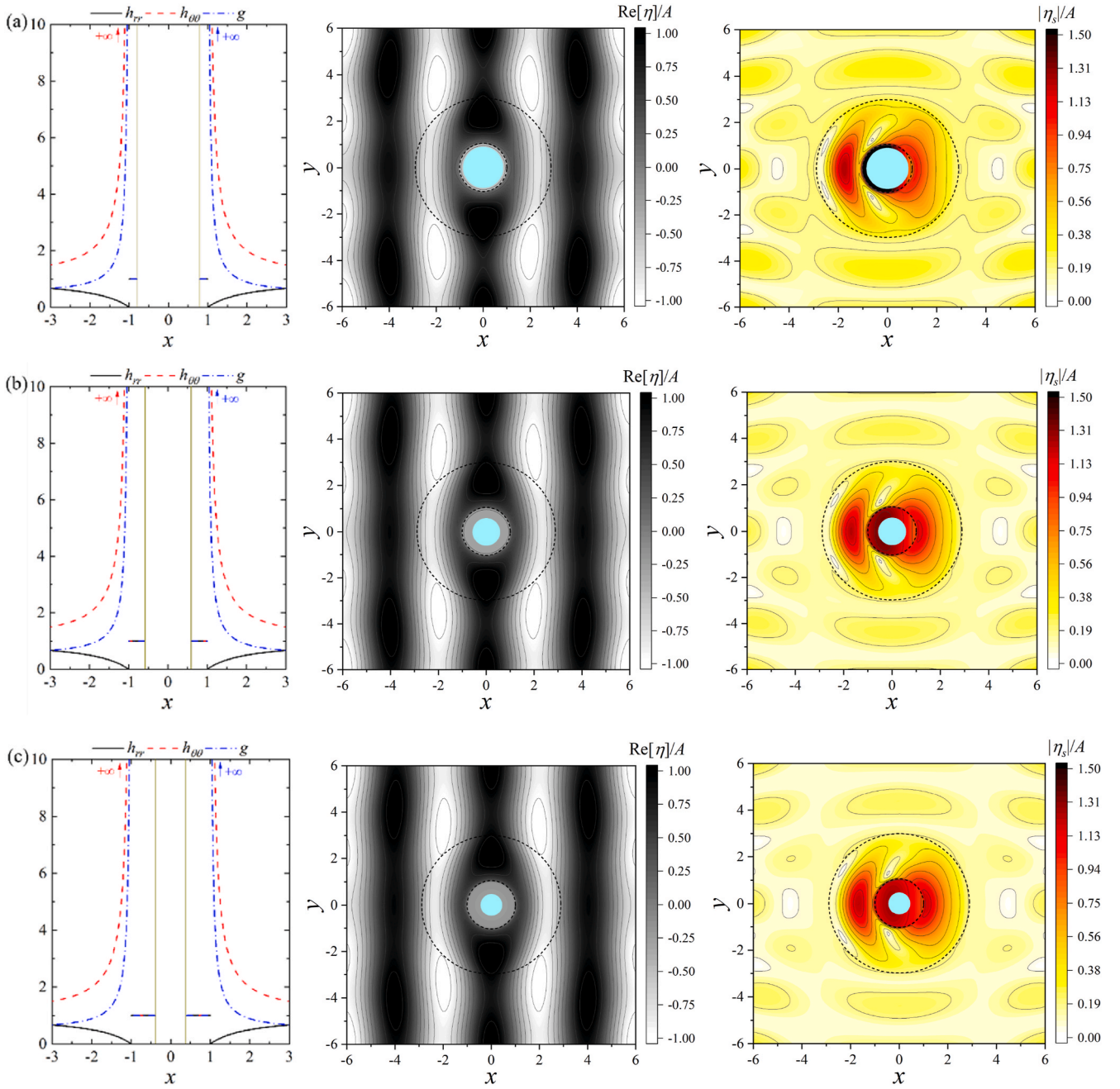


Fig. 5. Anisotropic medium parameters (h_{rr} , $h_{\theta\theta}$, and g), total-wave distribution ($\text{Re}[\eta]/A$), and scattered-wave distribution ($|\eta_s|/A$): (a) $r_0 = 0.8$ m; (b) $r_0 = 0.6$ m; (c) $r_0 = 0.4$ m; (d) $r_0 = 0.2$ m; (e) $r_0 = 0.0$ m.

wave directions. Here, the cylindrical coordinate system is used. Fig. 2 (a) represents the original space $\Omega(r, \theta)$ as a comparison reference, in which two circle lines $r = a$ and $r = b$ are marked. Seeing the red line in Fig. 2 (a), the wave path is a straight line. Fig. 2 (b) shows the transformed space $\Omega^*(r^*, \theta^*)$ of the annular cloak. It can be obtained by compressing the original space $0 < r < b$ into $a < r < b$. It can be seen that the wave propagates around the area $r < a$ due to space deformation. Assuming that there is a cylinder with a radius of a , the cylinder can be cloaked. However, the inner boundary $r = a$ is transformed from the origin point. Thus, the inner boundary should satisfy the no-flux boundary condition. Similar to that in Fig. 1, the anisotropic medium parameters can be obtained to realize the same wave field at the original space instead of using a transformed space, as shown in Fig. 2 (c).

A general linear transformation for the annular cloak is adopted. The

relationships between the original space Ω and the transformed space Ω^* are as follows,

$$\begin{cases} r^* = a + r(b-a)/b \\ \theta^* = \theta \end{cases} \quad (6)$$

According to Eq. (5), we can get the space scale factors $\tilde{h}$ and g shown in Fig. 2 (c),

$$\tilde{h} = \begin{bmatrix} \frac{r^* - a}{r^*} & 0 \\ 0 & \frac{r^*}{r^* - a} \end{bmatrix}, g = \left(\frac{b-a}{b}\right)^2 \frac{r^*}{r^* - a} \quad (7)$$

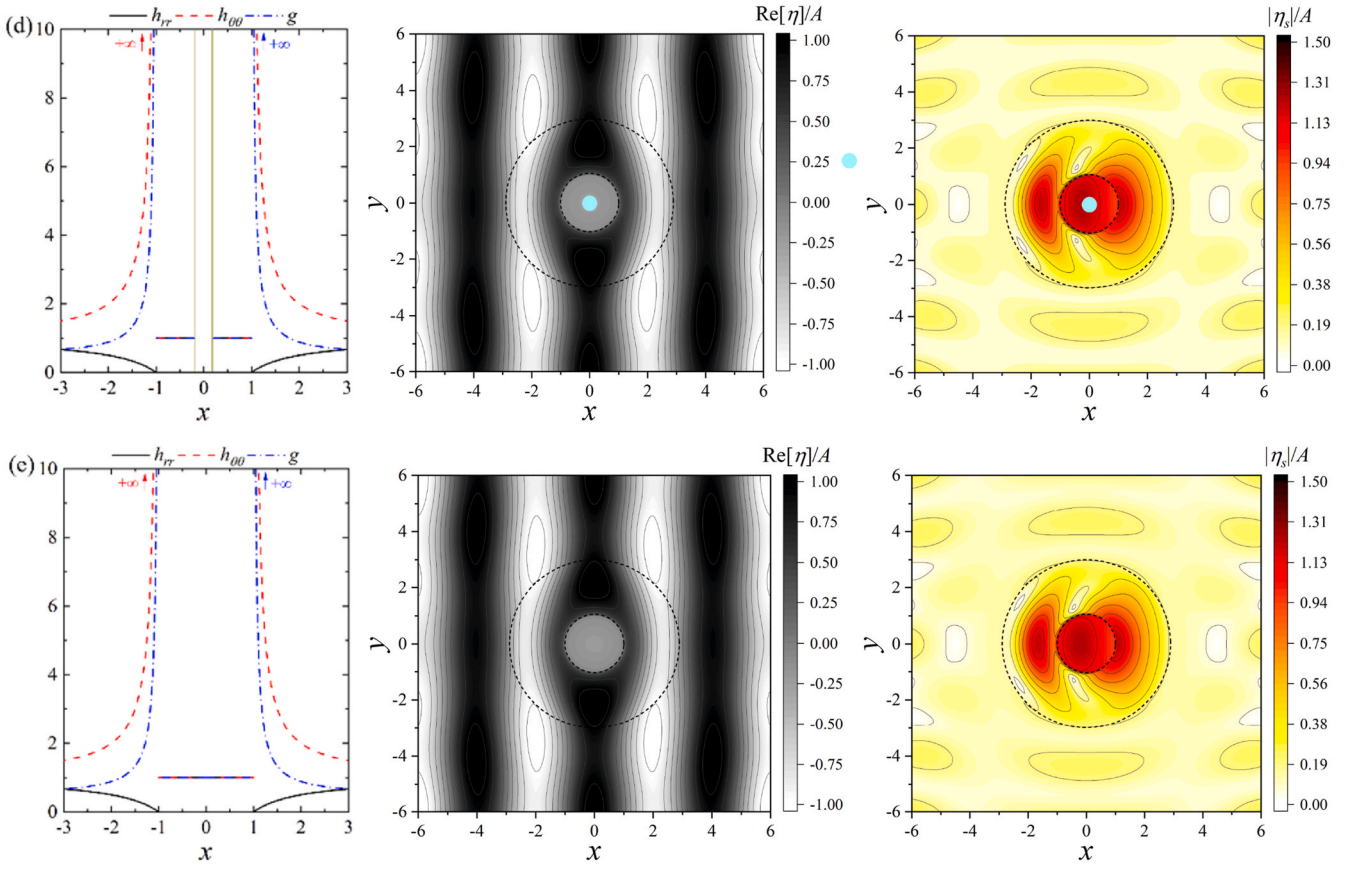
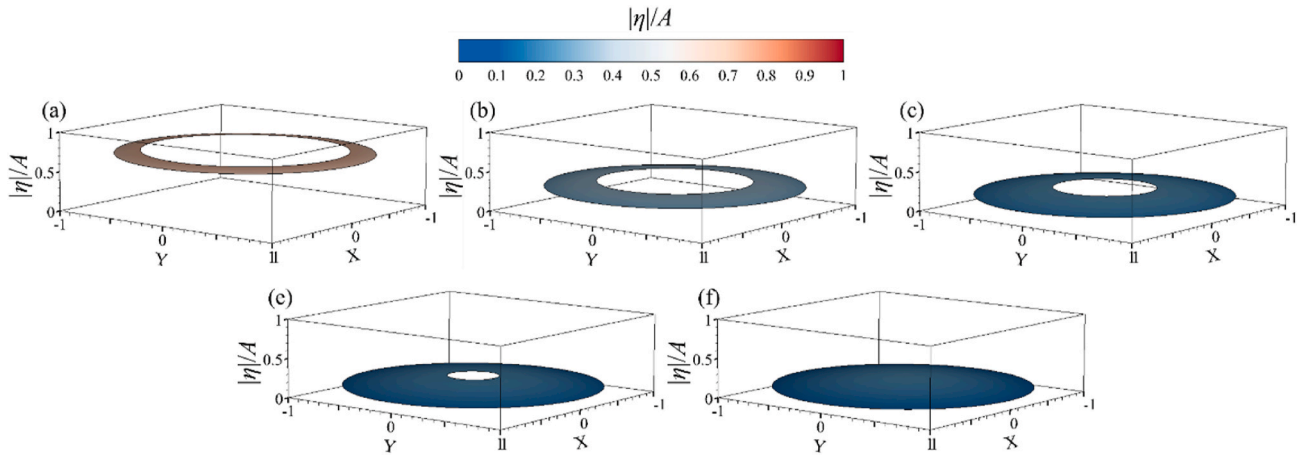


Fig. 5. (continued).

Fig. 6. Total-wave amplitudes ($|\eta|/A$): (a) $r_0 = 0.8$ m; (b) $r_0 = 0.6$ m; (c) $r_0 = 0.4$ m; (d) $r_0 = 0.2$ m; (e) $r_0 = 0.0$ m.

Here, $\tilde{h}$ is a tensor matrix $h_{rr} \neq h_{00}$, which means the water depth is anisotropic. If the scale factors were known, then the medium parameters ($\tilde{h}^* = \tilde{h}h_0$ and $g^* = gg_0$) can be obtained. Note that $\tilde{h} = 1.0$ and $g = 1.0$ at $r^* > b$. Based on Eq. (3), the same wave distribution in the transformed space [Fig. 2 (b)] can be realized by resetting the medium parameters ($\tilde{h}^*$ and g^*) in the original space [Fig. 2 (c)]. Here, the original space is the cylindrical coordinate space. Thus, we can drop the asterisk of r^* in Eq. (7) for aesthetic reasons (Schurig et al., 2006).

3. Results and discussion

The geometrical dimensions of the cloak and anisotropic medium parameters are determined by the values of a and b . The values of a and b are arbitrary and should satisfy $a < b$. In this study, $a = 1.0$ m and $b = 3.0$ m were adopted. Total- and scattered-wave distributions were analyzed by solving the Helmholtz equation to validate the STM-based cloak. The model setup for the annular cloak is shown in Fig. 3. A computational domain with a size of $50 \text{ m} \times 50 \text{ m}$ was adopted. As shown in Fig. 2, the medium parameters at $a < r < b$ can be calculated using Eq. (7).

The boundaries of this fluid region were defined by $S_{-\infty}$, S_{∞} , S_0 , and

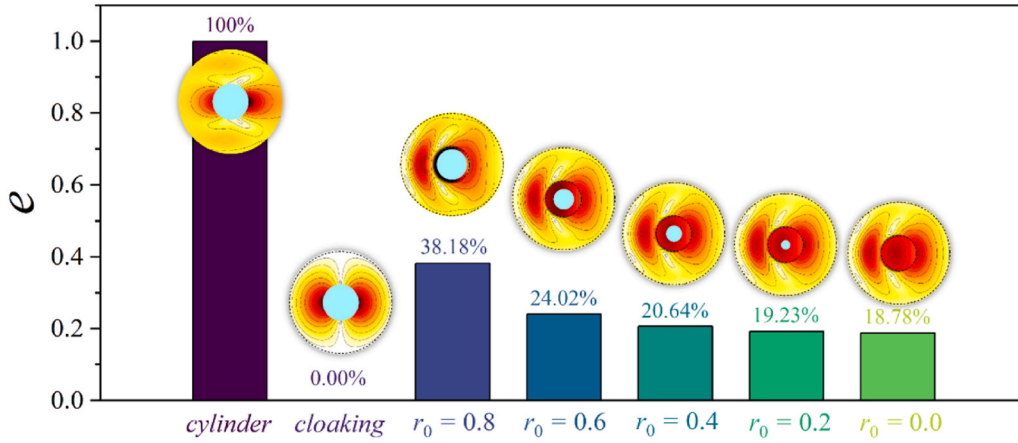


Fig. 7. Scattering attenuation coefficient e of single cylinder, cloaked cylinder, and $r_0 = 0.8, 0.6, 0.4, 0.2$, and 0.0 m.

S_H . The no-flux boundary conditions were applied at S_0 and S_H . The scattered boundary conditions were applied at $S_{-\infty}$ and S_{∞} . The Helmholtz equation and the boundary conditions are expressed by Eq. (8). $\tilde{h}$ and g are the scale factors, and $k_0 = \omega^2/g_0 h_0$ is the wavenumber. The incident wavenumber, $k_0 = \pi/2 \text{ m}^{-1}$, corresponding to the wavelength, $\lambda = 4.0 \text{ m}$, was considered. The incident wave ($\eta_0 = A e^{-ik_0 x}$) was generated at $S_{-\infty}$, and A is the wave amplitude, $A = 0.1 \text{ m}$.

$$\begin{cases} \nabla \cdot (\tilde{h} \nabla \eta) + \frac{k_0^2 \eta}{g} = 0, \text{ at } xoy \\ \frac{\partial \eta}{\partial n} = 0, \text{ on } S_H, S_0 \\ \frac{\partial \eta}{\partial n} = -ik\eta + 2ik\eta_0, \text{ on } S_{-\infty} \\ \frac{\partial \eta}{\partial n} = -ik\eta, \text{ on } S_{\infty} \end{cases} \quad (8)$$

To determine the scattering-cloaking efficiency, we used the scattering attenuation coefficient, e , defined as follows (Porter and Newman, 2014):

$$e = \frac{\Sigma_{\text{cloak}}}{\Sigma_{\text{cylinder}}} \quad (9)$$

where Σ_{cloak} and Σ_{cylinder} are the total scattering cross-sections (total energy scattered to infinity) of the cloaked and single cylinders, respectively.

The total scattering cross-sections of the cloaked cylinder (Σ_{cloak}) and single cylinder (Σ_{cylinder}) can be calculated as follows:

$$\Sigma = \frac{1}{2\pi} \int_0^{2\pi} |\eta_s(r, \theta)|^2 d\theta, r \rightarrow \infty \quad (10)$$

where η_s is the scattered-wave elevation, $\eta_s = \eta - \eta_0$.

Because of the limitations of the computational domain, we define a circle l far from the origin point to determine the scattering cross-section, as shown in Fig. 3. Here, $l = 15.0 \text{ m}$.

A finite element method based on COMSOL Multiphysics was used to solve the water wave problem. The total-wave distribution ($\text{Re}[\eta]/A$) and scattered-wave distribution ($|\eta_s|/A$) were calculated. Here, “ $\text{Re}[\]$ ” denotes taking the real part of the quantity in square brackets, whereas “ $| \ |$ ” represents the modulus of the quantity. The wave elevation of the total waves ($\text{Re}[\eta]/A$) and the wave amplitude of the scattered waves ($|\eta_s|/A$) were adopted to determine the cloaking efficiency. The cloaking efficiency can be identified by the regularity of the total-wave distribution ($\text{Re}[\eta]/A$) and the color depth of the scattered-wave distribution (energy view). For higher cloaking efficiency, the total-wave distribution should be more regular, and the scattered-wave distribution should

be lighter.

3.1. Validation of STM-based cloak

First, the wave distributions of a single cylinder with a radius of $r_0 = a = 1.0 \text{ m}$ (uncloaked body) and a cloaked cylinder with the anisotropic medium at $1.0 \text{ m} < r < 3.0 \text{ m}$ were determined, as shown in Fig. 4. In each subfigure, the left part represents the scale factors of the anisotropic medium parameters (h_{rr} , $h_{\theta\theta}$, and g) along the x -axis, the middle part represents the total-wave distribution ($\text{Re}[\eta]/A$), and the right part represents the scattered-wave distribution ($|\eta_s|/A$).

As shown in Fig. 4 (a), the vertical green solid lines in the left subfigure represent the cylinder boundaries. For the single cylinder, h_{rr} , $h_{\theta\theta}$, and g were equal to 1.0 at $1.0 \text{ m} < r < 3.0 \text{ m}$ because no anisotropic medium was present. Cylinder scattering, which results in disturbance of the wave fields, can be observed in the right subfigure. Fig. 4(b) shows the results for the cloaked cylinder. $r = b = 3.0 \text{ m}$ is plotted as a dashed line, which is the boundary of the anisotropic medium. The left subfigure shows that h_{rr} decreases with decreasing distance from the cylinder boundary, whereas $h_{\theta\theta}$ and g increase with decreasing distance from the cylinder boundary. $h_{\theta\theta}$ and g tended to be infinite at the area close to the cylinder boundary. With the anisotropic medium, almost no disturbance is observed in the middle subfigure, indicating that the cylinder was cloaked. The right subfigure shows that some scattered waves existed in the near field, although no scattered waves propagated into the far field. Scattered waves were generated by the anisotropic medium that changed the propagation path of the incident waves to avoid the space occupied by the cylinder. By applying Eq. (9), the scattering attenuation coefficient e of the cloak was calculated. The values of e were approximately zero, and the phenomenon could be regarded as perfect cloaking. These results validate the cloaking effects of an anisotropic medium.

3.2. Effects of no-flux inner boundary of anisotropic medium

Based on the design process of the annular cloak shown in Fig. 2, the inner boundary of the anisotropic medium ($r = a = 1.0 \text{ m}$) evolves from the origin. The inner boundary should be impervious and is physically regarded as a circular wall of radius a , as shown in Fig. 4(b). Thus, it is impossible to protect a freely moving object inside the wall, but only a fixed offshore structure, such as a wind turbine, can be protected. Five smaller cylinders with radii $r_0 = 0.8, 0.6, 0.4, 0.2$, and 0.0 m were selected to investigate the effects of the inner boundary locations on cloaking. $r_0 = 0.0 \text{ m}$ indicates that no cylinder was used. The distribution of the anisotropic medium at $1.0 \text{ m} < r < 3.0 \text{ m}$ did not change, whereas h_{rr} , $h_{\theta\theta}$, and g were equal to 1.0 at the area ($r_0 < r < 1.0 \text{ m}$) between the cylinder boundary and the inner boundary of the anisotropic medium.

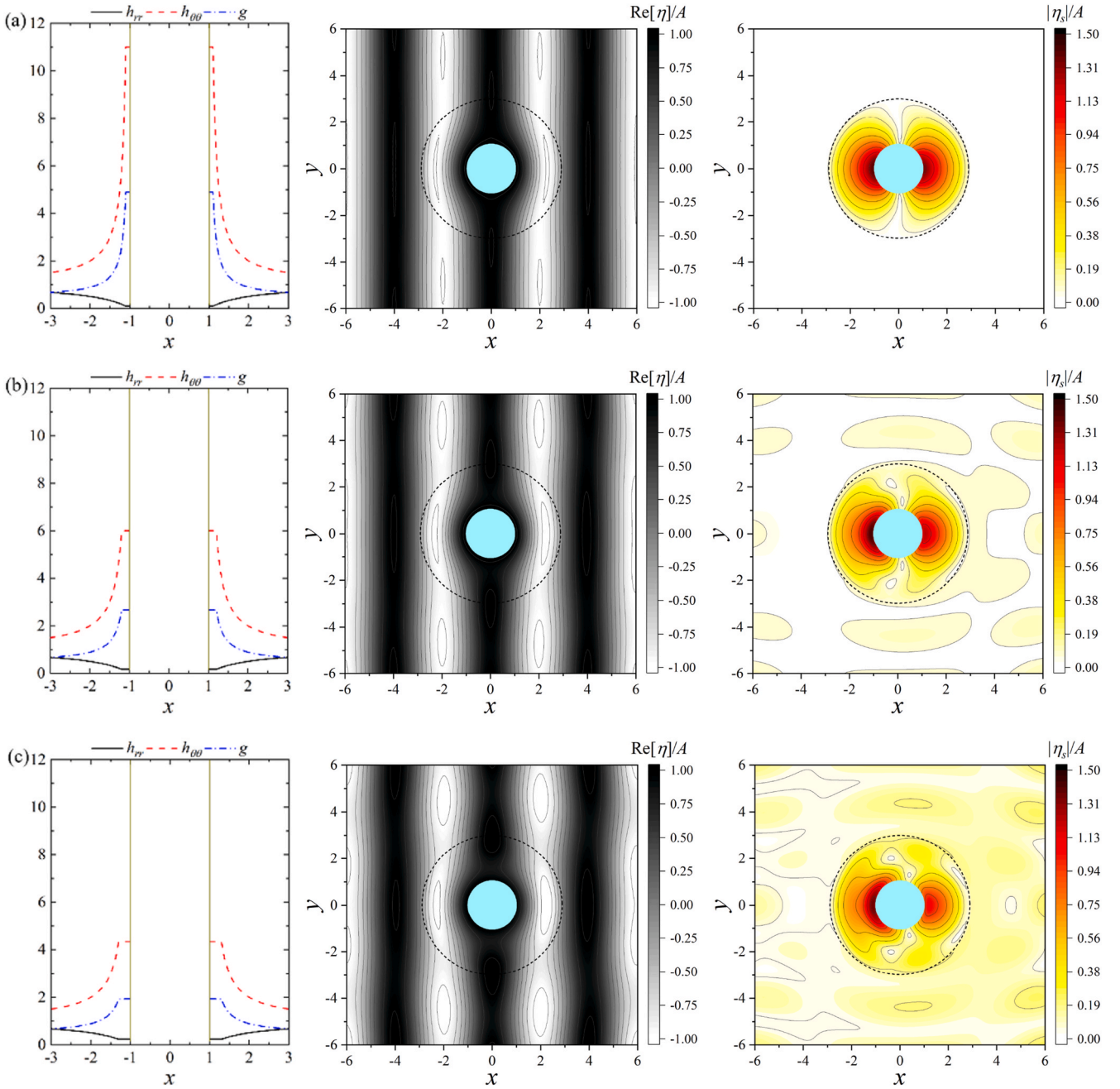


Fig. 8. Anisotropic medium parameters (h_{rr} , $h_{\theta\theta}$, and g), total-wave distribution ($\text{Re}[\eta]/A$), and scattered-wave distribution ($|\eta_s|/A$): (a) $r^0 = 1.1$ m; (b) $r^0 = 1.2$ m; (c) $r^0 = 1.3$ m; (d) $r^0 = 1.4$ m; (e) $r^0 = 1.5$ m.

Similar to parameters in Fig. 4, the anisotropic medium parameters (h_{rr} , $h_{\theta\theta}$, and g) along the x-axis, the total-wave distribution ($\text{Re}[\eta]/A$), and the scattered-wave distribution ($|\eta_s|/A$) of $r_0 = 0.8, 0.6, 0.4, 0.2$, and 0.0 m are presented in Fig. 5(a)–(e), respectively. An apparent disturbance of the total-wave field was observed at $r_0 = 0.8$ m, whereas significant scattered waves propagating to the far field were observed (Fig. 5(a)), indicating that the cloaking effects declined. As shown in Fig. 5(b)–(e), the total-wave fields tended to be regular with a decrease in the cylinder radius, whereas fewer scattered waves existed outside the cloaking area ($r > 3.0$ m), indicating that the cloaking effects improved compared to those of $r_0 = 0.8$ m.

The total-wave amplitudes ($|\eta|/A$) at the gap between the cylinder boundary and the inner boundary of the anisotropic medium ($r_0 < r < 1.0$ m) were calculated to evaluate the hydrodynamic protection of the

cylinder (Fig. 6). The results for $r_0 = 0.8, 0.6, 0.4, 0.2$, and 0.0 m are presented in Fig. 6(a)–(e), respectively. The dimensionless wave amplitudes were lower than 1.0, indicating that the wave energy around the cylinders were weaker than that around the incident waves. For $r_0 = 0.0$ m (Fig. 6(e)), the maximum wave amplitude was $|\eta|/A = 0.26$, which exhibited a decrease of 74% compared to that of the incident waves. This implies that a ship can be positioned in an inner area for protection without a wall. Although the no-flux boundary moved inward, the cylinder was somewhat hydrodynamically protected. The scattering attenuation coefficient e of $r_0 = 0.8, 0.6, 0.4, 0.2$, and 0.0 m were calculated to further analyze the cloaking effects of various inner boundary locations quantitatively (Fig. 7). The corresponding scattered-wave distribution around the cylinders in each case are shown in Fig. 7. The scattering attenuation coefficients e of the single and cloaked

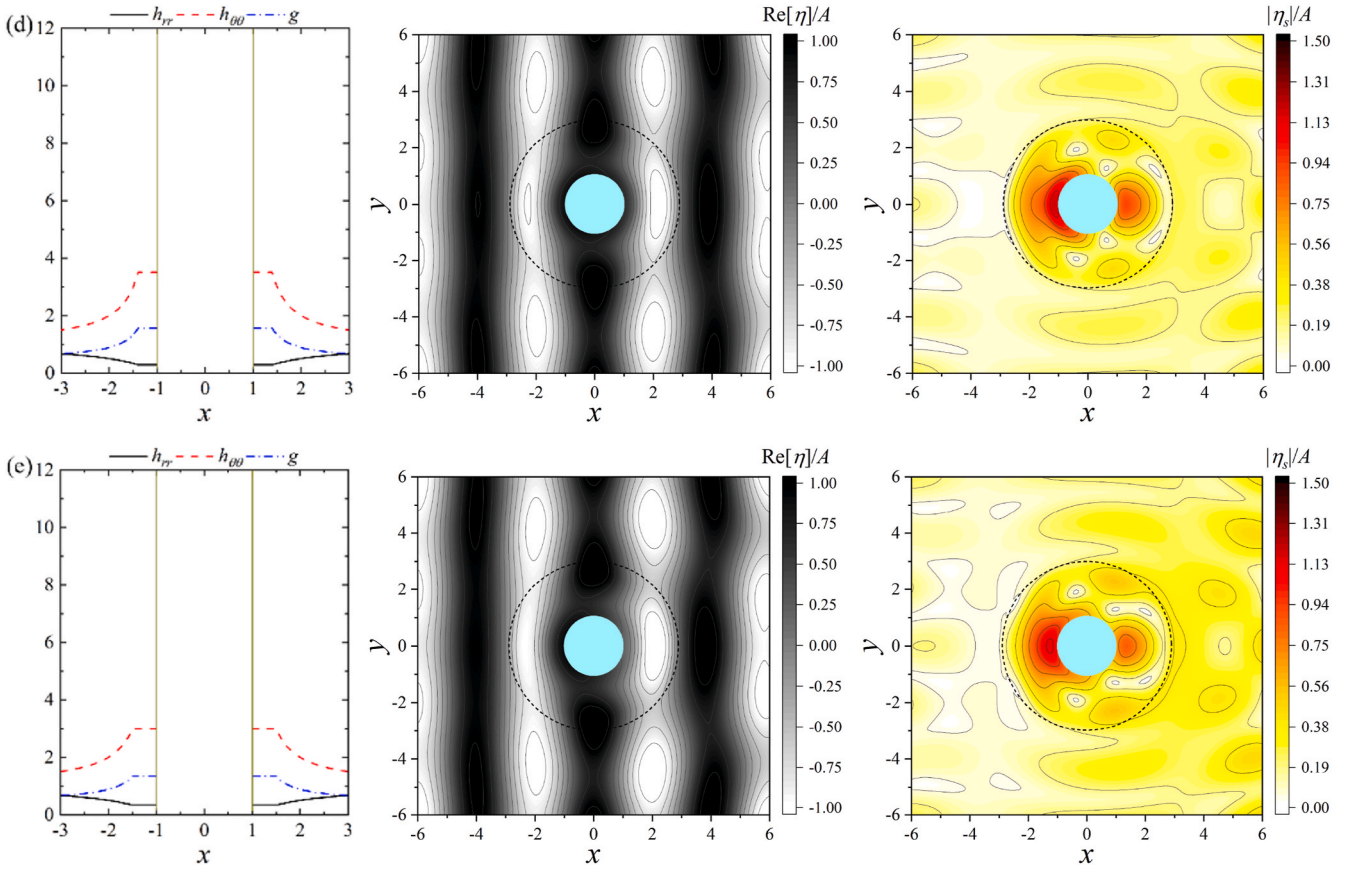
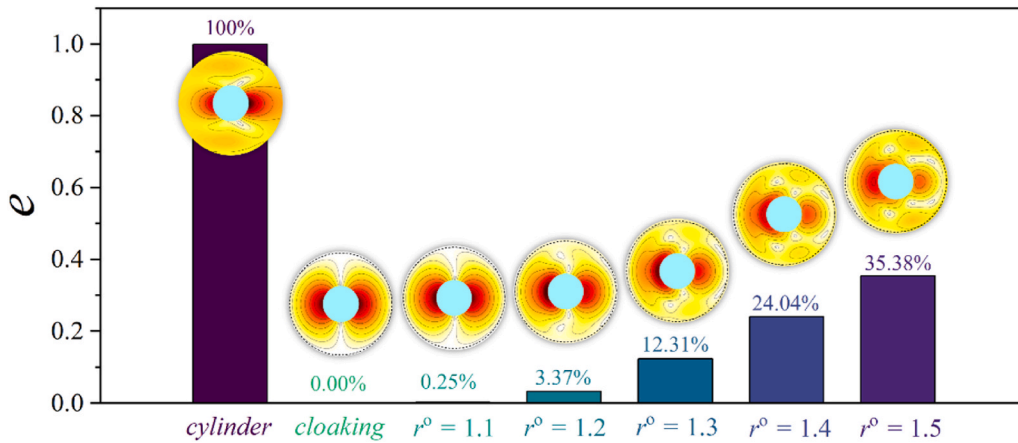


Fig. 8. (continued).

Fig. 9. Scattering attenuation coefficients e of single cylinder, cloaked cylinder, and $r^0 = 1.1, 1.2, 1.3, 1.4$, and 1.5 m.

cylinders are shown as references. The scattering attenuation coefficient e decreased with decreasing cylinder radius. In the case of no cylinder ($r_0 = 0.0$ m), more than 80% of the scattered-wave energy was eliminated. This means that an anisotropic medium can efficiently bend incident waves to avoid areas without a no-flux boundary.

3.3. Simplification of anisotropic medium parameters

The medium parameters, $h_{\theta\theta}$ and g , were singular at the boundary of $r = a = 1.0$ m, as shown in Fig. 4(b). A simplified method known as numerical truncation is proposed in this study to solve this problem. Specifically, the values of the three parameters ($h_{rr}(r)$, $h_{\theta\theta}(r)$, and $g(r)$)

at $1.0 \text{ m} < r < r^0$ were selected as their values at $r = r^0$ ($h_{rr}(r^0)$, $h_{\theta\theta}(r^0)$, $g(r^0)$), respectively. The radius of the cylinder did not change. Five cases ($r^0 = 1.1, 1.2, 1.3, 1.4$, and 1.5 m) were considered to investigate the effects of this simplification method on cloaking. The anisotropic medium parameters (h_{rr} , $h_{\theta\theta}$, and g) along the x -axis, the total-wave distribution ($\text{Re}[\eta]/A$), and the scattered-wave distribution ($|\eta_s|/A$) of $r^0 = 1.1, 1.2, 1.3, 1.4$, and 1.5 m are shown in Fig. 8(a)–(e), respectively. The left subfigure in Fig. 8(a) shows that $h_{rr} = 0.09$, $h_{\theta\theta} = 11.00$, and $g = 4.89$ are adopted at $1.0 \text{ m} < r < 1.1$ m. Cloaking was almost unaffected by parameter simplification, as shown in the middle and right subfigures in Fig. 8(a). Fig. 8(b)–(e) show that the cloaking effects are weakened with an increase in r^0 . However, the anisotropic medium could still bend the

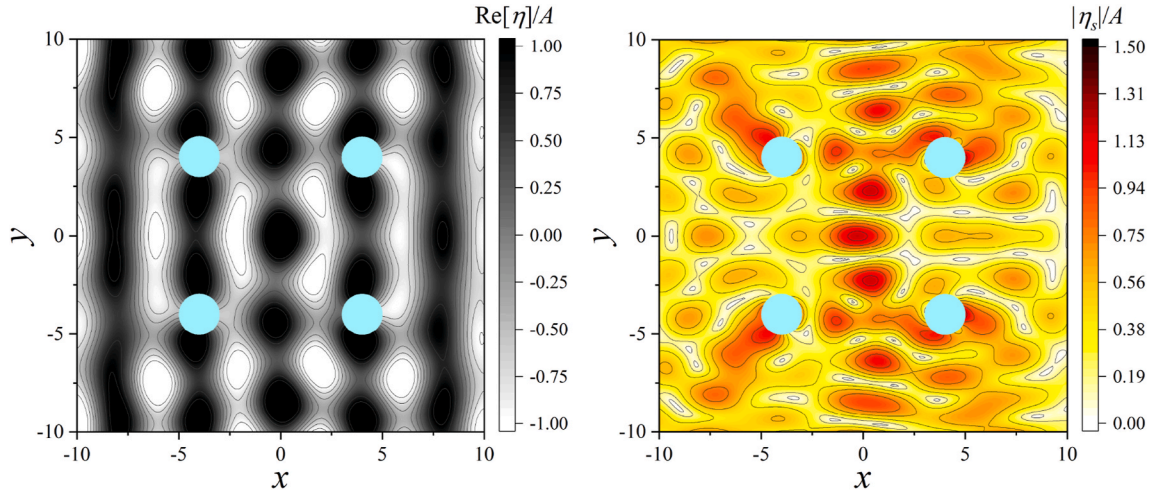


Fig. 10. The total-wave distribution ($\text{Re}[\eta]/A$), and the scattered wave distribution ($|\eta_s|/A$) of the uncloaked four-column structure.

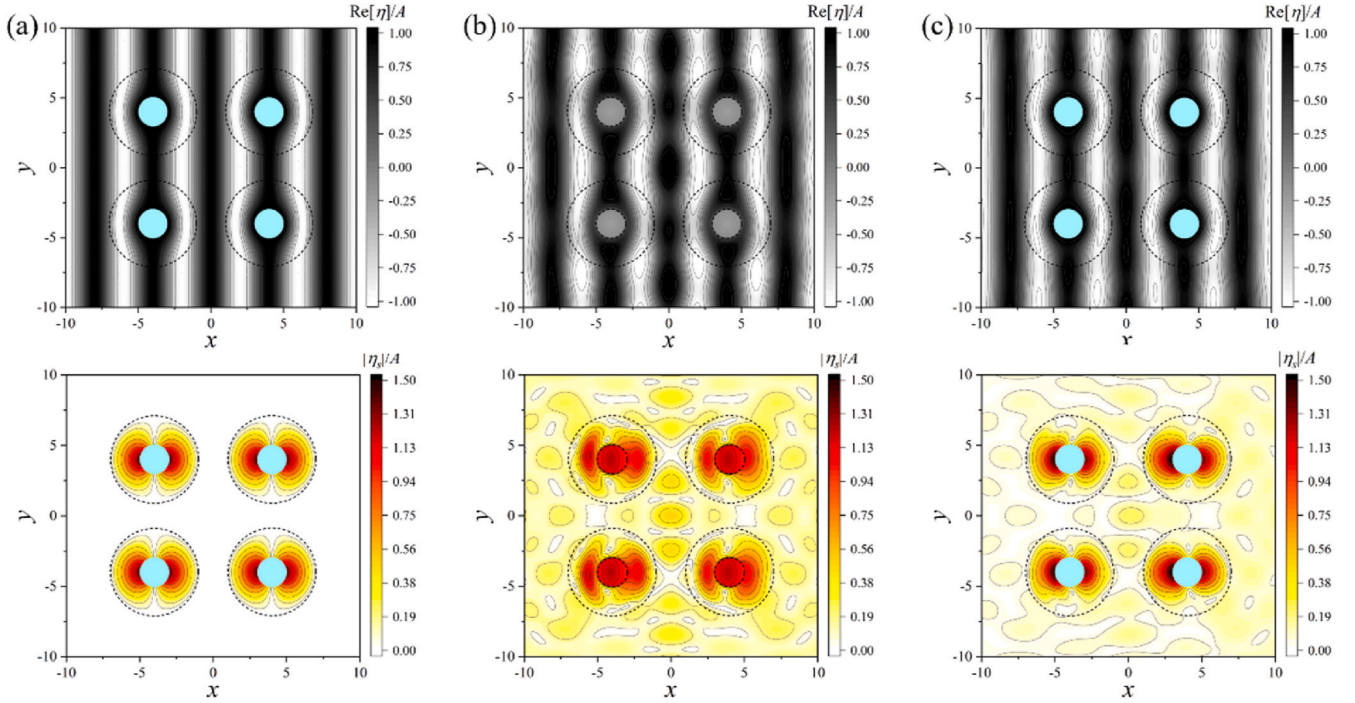


Fig. 11. The total-wave distribution ($\text{Re}[\eta]/A$), and the scattered wave distribution ($|\eta_s|/A$) of the cloaked four-column structure: (a) perfect cloak; (b) cloak without no-flux boundary of $r_0(i) = 0.0\text{m}$; (c) cloak with simplified parameters of $r^0(i) = 1.2\text{ m}$.

incident wave path at $1.0\text{ m} < r < 3.0\text{ m}$ to protect the cylinder, even for $r^0 = 1.5\text{ m}$, as shown in Fig. 8(e).

The scattering attenuation coefficients e of $r^0 = 1.1, 1.2, 1.3, 1.4$, and 1.5 m were calculated to further analyze the cloaking effects of different parameter simplifications quantitatively (Fig. 9). The scattering attenuation coefficients e of the single and cloaked cylinders are shown as references. The scattering attenuation coefficient e increased with increasing r^0 . In the case of $r^0 = 1.1\text{ m}$, more than 99% of the scattered-wave energy was eliminated. For $r^0 = 1.5\text{ m}$, more than 64% of the scattered-wave energy was eliminated, indicating that the proposed simplified anisotropic medium effectively protects structures exposed to water waves.

3.4. Cloaking multi-bodies using anisotropic medium

The anisotropic medium has been validated to cloak a cylinder in

water waves. This study introduced the anisotropic medium to cloak a four-column structure. To obtain the anisotropic medium parameters (h_{rr} , $h_{\theta\theta}$, and g), the local coordinate systems r_i - θ_i ($i = 1, 2, 3, 4$) located at the center of each column were considered. Note that r_i should be used in Eq. (7) to calculate the anisotropic medium parameters for each column. The other model set of this four-column structure is similar to that of the single cylinder.

First, the wave distributions of the uncloaked four-column structure were calculated, as shown in Fig. 10. In Fig. 10, the left subfigure shows the total-wave distribution ($\text{Re}[\eta]/A$); the right one represents the scattered-wave distribution ($|\eta_s|/A$). Obvious disturbance of the total-wave fields can be found due to the scattered waves generated by the four columns. Then, the wave distributions of the cloaked multi-bodies with the perfect anisotropic medium were calculated as shown in Fig. 11 (a). The distribution of the three parameters in each local coordinate system is the same as that shown in Fig. 4 (b). According to the total-

wave distribution, it can be found the perfect cloaked four-column structure does not disturb the incident wave field. It can also be found that the scattered waves generated by the anisotropic medium of each cylinder bend the incident wave path to cloak the cylinder, which means the cylinders can not affect each other. Next, the wave distribution of the case without no-flux boundary ($r_0(i) = 0.0$ m) was calculated, as shown in Fig. 11 (b). It can be found that the wave distributions around each area are similar to those in Fig. 5 (e). Besides, the cloaking effects of the four-column structure with the simplified anisotropic medium parameters ($r^0(i) = 1.2$ m) are studied, as shown in Fig. 11 (c). It can be found that the wave distributions around each column are similar to that in Fig. 8 (b). Thus, the simplified anisotropic medium can also hydrodynamically protect the multi-bodies structure in water waves. Note that there should be no overlap areas between anisotropic media. If several objects are close together, then they can be considered as one object to be cloaked with an anisotropic medium.

4. Conclusions

An anisotropic medium designed using the STM can cloak and hydrodynamically protect offshore structures from shallow-water waves. In this study, the anisotropic medium parameters (h_{rr} , $h_{\theta\theta}$, and g) of an annular cloak ($1.0 \text{ m} < r < 3.0 \text{ m}$) were designed. By investigating the total-wave distribution ($\text{Re}[\eta]/A$) and scattered-wave distribution ($|\eta_s|/A$), the STM-based cloak was validated. Subsequently, the effects of the inner boundary of the anisotropic medium on the cloaking efficiency were analyzed. Furthermore, a simplification method for the anisotropic medium parameters was developed and validated to demonstrate the cloaking effects. Finally, an anisotropic medium was introduced to cloak the four-column structure. The main conclusions of this study are as follows.

- 1) An annular cloak was designed using the STM in which the medium was anisotropic. By investigating the wave distribution around the cylinder, it was found that a perfect cloak could be realized using an anisotropic medium. Additionally, the scattered waves generated by the anisotropic medium bent the incident wave path to avoid the area where the body was located.
- 2) For a perfect cloak, the inner boundary of the anisotropic medium must satisfy no-flux boundary conditions. The effects of the inner boundary of the anisotropic medium on the cloaking efficiency were investigated by considering five smaller cylinders with the radii $r_0 = 0.8, 0.6, 0.4, 0.2$, and 0.0 m. The cloaking efficiency was found to be higher for smaller cylinders. The inside area could be hydrodynamically protected by an anisotropic medium, although the no-flux boundary did not exist.
- 3) The medium parameters tended to be infinite at the inner boundary of the anisotropic medium. Thus, a simplification method known as numerical truncation was developed. The values of the medium parameters (h_{rr} , $h_{\theta\theta}$, and g) at $1.0 \text{ m} < r < r^0$ were simplified to their values at $r = r^0$. Five cases ($r^0 = 1.1, 1.2, 1.3, 1.4$, and 1.5 m) were considered to examine the effects of this simplification method on cloaking. The results showed that the cloaking efficiency decreased with increasing r^0 . An anisotropic medium with simplified parameters is effective for cloaking a cylinder to an extent.
- 4) An anisotropic medium was introduced to cloak a four-column structure. Four annular cloaks were obtained by designing an anisotropic medium in the local coordinate system of each column. By comparing the wave distributions of the uncloaked four-column structure with those of the cloaked structure, the anisotropic medium was validated for cloaking the multibody structure exposed to water waves.

The designed anisotropic medium does not exist naturally. Future work should include the construction of an equivalent metamaterial to simulate the anisotropic medium. The proposed multibody cloak is

frequency-independent and omnidirectional and has wide application prospects in the hydrodynamic protection of offshore structures.

CRediT authorship contribution statement

Zhigang Zhang: Conceptualization, Investigation, Methodology, Software, Visualization, Writing – original draft. **Takahito Iida:** Formal analysis, Writing – review & editing. **Guanghua He:** Data curation, Formal analysis, Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing. **Weijie Mo:** Methodology, Writing – review & editing. **Shuang Liu:** Data curation, Investigation, Validation, Visualization. **Zhengxiao Luan:** Data curation, Formal analysis. **Penglin Jing:** Data curation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This study was supported by the China Postdoctoral Science Foundation (2023M742157), the Shandong Province Taishan Scholars Project (tsqn201909172), the Shandong Provincial Natural Science Foundation (ZR2023QA097), the Open Research Fund Program of State Key Laboratory of Coastal and Offshore Engineering, Dalian University of Technology (LP2309), the Innovation Project of Qingdao Post-Doctoral (QDBSH20230202121), and the China Scholarship Council (201906120145).

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